Chapter2

Ricky Dai

September 2026

1 Finite, Countable and Uncountable Sets

Def 1. Consider two sets A and B, and suppose that with each element of A there is associated, in some manner, an element of B, which we denote by $f(x)$. Then f is said to be a *function* from A to B (or a *mapping* of A into B). The set A is called the *domain* of f (we also say f is defined on A) and the elements $f(x)$ are called the *values* of f. The set of all values of f is called the *range* of f.

Def 2. Let A and B be two sets and let f be a mapping of A into B. If $E \subset A$, $f(E)$ is defined to be the set of all elements $f(x)$, for $x \in E$. We call $f(E)$ the *image* of E under f. In this notation, $f(A)$ is the range of f. It is clear that $f(A) \subset B$. If $f(A) = B$, we say that f maps A onto B.

Def 3. If there exists a $1-1$ mapping of A *onto* B, we say that A and B can be put in $1-1$ *correspondence*, or that A and B have the same *cardinal number*, or, briefly, that A and B are *equivalent*, and we write $A \sim B$. This relation clearly has the following properties:

Reflexive: $A \sim A$.

Symmetric: If $A \sim B$, then $B \sim A$.

Transitive: If $A \sim B$ and $B \sim C$, then $A \sim C$.

Any relation with these three properties is called an *equivalence relation*.

Def 4. For any positive n, let $\mathbb{N}_n$ be the set whose elements are the integers $1, 2, ..., n$; let $\mathbb{N}$ be the set consisting of all positive integers. For any set A, we say:

(a) A is finite if $A \sim \mathbb{N}_n$ for some n (the empty set is also considered to be finite).

(b) A is *infinite* if A is not finite.

(c) A is *countable* if $A \sim \mathbb{N}$.

(d) A is *uncountable* if A is neither finite nor countable.

(e) A is *at most countable* if A is finite or countable.

Countable sets are sometimes called *enumerable*, or *denumerable*.

Rmk 5. In the definition above, (b) can be replaced by the statement: A is infinite if A is equivalent to one of its proper subsets.

Def 6. By a *sequence*, we mean a function f defined on the set $\mathbb{N}$ of all positive integers. If $f(n) = x_n$, for $n \in \mathbb{N}$, it is customary to denote the sequence f by the symbol $(x_n)_{n=1}^{\infty}$, or more briefly by (x_n), or sometimes by $x_1, x_2, x_3, ...$ The values of f, that is, the elements x_n, are called the *terms* of the sequence. If A is a set and if $x_n \in A$ for all $n \in \mathbb{N}$, then (x_n) is said to be a *sequence in* A, or a *sequence of elements of* A.

Thm 7. Every infinite subset of a countable set A is countable.

Def 8. Let A and Ω be sets, and suppose that with each element α of A there is associated a subset of Ω which we denote by E_α.

The set whose elements are the sets E_α will be denoted by $\{E_\alpha\}$.

Rmk 9. communicative law: $A \bigcup B = B \bigcup A$; $A \bigcap B = B \bigcap A$.

associative law: $(A \bigcup B) \bigcup C = A \bigcup (B \bigcup C)$; $(A \bigcap B) \bigcap C = A \bigcap (B \bigcap C)$.

distribution law: $A \bigcap (B \bigcup C) = (A \bigcap B) \bigcup (A \bigcap C)$.

Thm 10. Let $\{E_n\}$, $n = 1, 2, 3, ...$, be a countable collection of countable sets, and put

$$S = \bigcup_{n=1}^{\infty} E_n.$$

Then S is countable.

Cor 11. Suppose A is at most countable, and, for every $\alpha \in A$, B_α is at most countable. Put

$$T = \bigcup_{\alpha \in A} B_\alpha.$$

Then T is at most countable, for T is equivalent to a subset of S as stated above.

Thm 12. Let A be a countable set, and let B_n be the set of all $n-tuples(a_1, ..., a_n)$, where $a_k \in A$ $(k = 1, ..., n)$, and the elements $a_1, ..., a_n$ need not be distinct. Then B_n is countable.

Cor 13. The set of all rational numbers is countable.

Thm 14. Let A be the set of all sequences whose elements are the digits 0 and 1. This set A is uncountable.

2 Metric Spaces

Def 15. A set X, whose elements we shall call *points*, is said to be a *metric space* if with any two points p and q of X there is associated a real number $d(p,q)$, called the *distance* from p to q, such that

(a) $d(p,q) > 0$ if $p \neq q$; $d(p,p) = 0$;

(b) $d(p,q) = d(q,p)$;

(c) $d(p,q) \leq d(p,r) + d(r,q)$ for any $r \in X$.

Any function with these three properties is called a *distance function*, or a *metric*.

Def 16. By the *segment* (a,b) we mean the set of all real numbers x such that $a < x < b$.

By the *interval* $[a,b]$ we mean the set of all real numbers x such that $a \leq x \leq b$.

If $a_i < b_i$ for $i = 1, ..., k$, the set of all points $\mathbf{x} = (x_1, ...x_k)$ in $\mathbb{R}^k$ whose coordinates satisfy the inequalities $a_i \leq x \leq b_i$ $(1 \leq i \leq k)$ is called a *k-cell*. Thus a 1-cell is an interval, a 2-cell is a rectangle, etc.

If $\mathbf{x} \in \mathbb{R}^k$ and $r > 0$, the *open (or closed) ball* B with center at $\mathbf{x}$ and radius r is defined to be the set of all $\mathbf{y} \in \mathbb{R}^k$ such that $|\mathbf{y} - \mathbf{x}| < r$ (or $|\mathbf{y} - \mathbf{x}| \leq r$).

We call a set $E \subset \mathbb{R}^k$ *convex* if

$$\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in E$$

whenever $\mathbf{x} \in E$, $\mathbf{y} \in E$, and $0 < \lambda < 1$.

Def 17. Let X be a metric space. All points and sets mentioned below are understood to be elements and subsets of X.

(a) *A neighborhood* of p is a set $N_r(p)$ consisting of all q such that $d(p,q) < r$ for some $r > 0$. The number r is called the *radius* of $N_r(p)$.

(b) A point p is a *limit point* of the set E if *every* neighborhood of p contains a point $q \neq p$ such that $q \in E$.

(c) If $p \in E$ and p is not a limit point of E, then p is called an *isolated point* of E.

(d) E is *closed* if every limit point of E is a point of E.

(e) A point p is an *interior* point of E if there is a neighborhood N of p such that $N \subset E$.

(f) E is *open* if every point of E is an interior point of E.

(g) The *complement* of E (denoted by E^c) is the set of all points $p \in X$ such that $p \notin E$.

(h) E is *perfect* if E is closed and if every point of E is a limit point of E.

(i) E is *bounded* if there is a real number M and a point $q \in X$ such that $d(p,q) < M$ for all $p \in E$.

(j) E is *dense in* X if every point of X is a limit point of E, or a point of E (or both).

Thm 18. Every neighborhood is an open set.

Thm 19. If p is a limit point if a set E, then every neighborhood of p contains infinitely many points of E.

Cor 20. A finite point set has no limit points.

Thm 21. Let $\{E_\alpha\}$ be a (finite or infinite) collection of sets E_α. Then

$$(\bigcup_\alpha E_\alpha)^c = \bigcap_\alpha (E_\alpha^c).$$

Thm 22. A set E is open if and only if its complement is closed.

Cor 23. A set F is closed if and only if its complement is open.

Thm 24. (a) For any collection $\{G_\alpha\}$ of open sets, $\bigcup_\alpha G_\alpha$ is open.

(b) For any collection $\{F_\alpha\}$ of closed sets, $\bigcap_\alpha F_\alpha$ is closed.

(c) For any finite collection $G_1, ..., G_n$ of open sets, $\bigcup_{i=1}^n G_i$ is open.

(d) For any finite collection $F_1, ..., F_n$ of closed sets, $\bigcap_{i=1}^n F_i$ is closed.

Def 25. If X is a metric space, if $E \subset X$, and if E' denotes the set of all limit points of E in X, then the *closure* of E is the set $\overline{E} = E \bigcup E'$.

Thm 26. If X is a metric space and $E \subset X$, then

(a) $\overline{E}$ is closed,

(b) $E = \overline{E}$ if and only if E is closed,

(c) $\overline{E} \subset F$ for every closed set $F \subset X$ such that $E \subset F$.

Thm 27. Let E be a nonempty set of real numbers which is bounded above. Let $y = supE$. Then $y \in \overline{E}$. Hence $y \in E$ if E is closed.

Thm 28. Suppose $Y \subset X$. A subset E of Y is open relative to Y if and only if $E = Y \bigcap G$ for some open subset G of X.

3 Compact Sets

Def 29. By an *open cover* of a set E in a metric space X we mean a collection $\{G_\alpha\}$ of open subset of X such that $E \subset \bigcup_\alpha G_\alpha$.

Def 30. A subset K of a metric space X is said to be *compact* if every open cover of K contains a *finite* subcover.

Thm 31. Suppose $K \subset Y \subset X$. Then K is compact relative to X if and only if K is compact relative to Y.

Thm 32. Compact subsets of metric spaces are closed.

Thm 33. Closed subsets of compact sets are compact.

Cor 34. If F is closed and K is compact, then $F \bigcap K$ is compact.

Thm 35. If $\{K_\alpha\}$ is a collection of compact subsets of a metric space X such that the intersection of every finite subcollection of $\{K_\alpha\}$ is nonempty, then $\bigcap K_\alpha$ is nonempty.

Cor 36. If $\{K_n\}$ is a countable collection of nonempty compact sets such that $K_n \supset K_{n+1}$ $(n = 1, 2, 3, ...)$, then $\bigcap_{n=1}^{\infty} K_n$ is not empty

Thm 37. If E is an infinite subset of a compact set K, then E has a limit point in K.

Thm 38. If $\{I_n\}$ is a sequence of intervals in $\mathbb{R}^1$, such that $I_n \supset I_{n+1}$ $(n = 1, 2, 3, ...)$, then $\bigcap_{n=1}^{\infty} I_n$ is not empty.

Thm 39. Let k be a positive integer. If $\{I_n\}$ is a sequence of k-cells such that $I_n \supset I_{n+1}$ $(n = 1, 2, 3, ...)$, then $\bigcap_{n=1}^{\infty} I_n$ is not empty.

Thm 40. Every k-cell is compact.

Thm 41. If a set E in $\mathbb{R}^k$ has one of the following three properties, then it has the other two:

(a) E is closed and bounded.

(b) E is compact.

(c) Every infinite subset of E has a limit point in E.

Thm 42. (Weierstrass). Every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$.

4 Perfect Sets

Thm 43. Let P be a nonempty perfect set in $\mathbb{R}^k$. Then P is uncountable.

Cor 44. Every interval $[a, b]$ $(a < b)$ is uncountable. In particular, the set of all real numbers is uncountable.

Def 45. The Cantor Set. The set which we are now going to construct shows that there exist perfect sets in R^1 which contain no segment.

Let E_0 be the interval $[0, 1]$. Remove the segment $(1/3, 2/3)$, and let E_1 be the union of the intervals

$$[0, \frac{1}{3}], [\frac{2}{3}, 1].$$

Remove the middle thirds of these intervals, and let E_2 be the union of the intervals

$$[0, \frac{1}{9}], [\frac{2}{9}, \frac{3}{9}], [\frac{6}{9}, \frac{7}{9}], [\frac{8}{9}, 1]$$

Continuing in the way, we obtain a sequence of compact sets E_n such that

(a) $E_1 \supset E_2 \supset E_3 \supset ...$;

(b) E_n is the union of 2^n intervals, each of length 3^{-n}.

The set

$$P = \bigcap_{n=1}^{\infty} E_n$$

is called the *Cantor set.* P is clearly compact and nonempty.

5 Connected Sets

Def 46. Two subsets A and B of a metric space X are said to be *separated* if both $A \bigcap \overline{B}$ and $\overline{A} \bigcap B$ are empty, i.e., if no point of A lies in the closure of B and no point of B lies in the closure of A.

A set $E \subset X$ os said to be *connected* if E is *not* a union of two nonempty separated sets.

Rmk 47. Separated sets are of course disjoint, but disjoint sets need not be separated. For example, the interval $[0, 1]$ and the segment $(1, 2)$ are *not* separated since 1 is a limit point of $(1, 2)$. However, the segments $(0, 1)$ and $(1, 2)$ *are* separated.

Thm 48. A subset E of the real line $\mathbb{R}^1$ is connected if and only if it has the following property: If $x \in E$, $y \in E$, and $x < z < y$, then $z \in E$.

6 Exercise Notes

Thm 49. All algebraic numbers is countable.

Cor 50. All transcendental numbers is uncountable.

Def 51. Let E° denote the set of all interior points of a set E¿ Then E° is called the interior of E.

Thm 52. Every connected metric space with at least two points is uncountable.

Def 53. A metric space is called *separable* if it contains a countable dense subset.

Def 54. A collection $\{V_\alpha\}$ of open subsets of X is said to be a *base* for X if the following is true: For every $x \in X$ and every open set $G \subset X$ such that $x \in G$, we have $x \in V_\alpha \subset G$ for some α. In other words, every open set in X is the union of a subcollection of $\{V_\alpha\}$.

Thm 55. Every separable metric space has a *countable* base.

Thm 56. If a metric space has a countable base, then it is separable.

Cor 57. For a metric space, having a countable base $\iff$ being separable.

Thm 58. Every compact metric space K has a countable base, and that K is therefore separable.

Thm 59. Let X be a metric space in which every infinite subset has a limit point. Then X is compact.

Def 60. A point p in a metric space X is a *condensation point* of a set $E \subset X$ if every neighborhood of p contains uncountably many points of E.

Thm 61. Suppose $E \subset \mathbb{R}^k$, E is uncountable, and let P be the set of all condensation points of E. Then P is perfect and that at most countably many points of E are not in P.

Thm 62. If $\mathbb{R}^k = \bigcup_{n=1}^\infty F_n$, where each F_n is a closed subset of $\mathbb{R}^k$, then at least one F_n has a nonempty interior.

Equivalent statement: If $G_n (= \mathbb{R}^k \setminus F_n)$ is a dense open subset of $\mathbb{R}^k$ for $n = 1, 2, 3, ...$, then $\bigcap_{n=1}^\infty G_n$ is not empty.

Rmk 63. Relations between Some Concepts:

Compact	$\Longleftrightarrow$	Closed + Bounded $\Longleftrightarrow$ Every infinite subset has a limit point in the set;
Compact	$\Longrightarrow$	Separable;
Compact	$\Longrightarrow$	Every open cover contains a *finite* subcover;
Separable	$\Longleftrightarrow$	Having a countable base;
Perfect	$\Longleftrightarrow$	Closed + No isolated points (or say every point is a limit point);
Cantor Set	$\Longrightarrow$	Compact and Perfect.